Quality Ladder / Schumpeterian Growth Cheatsheet

Macroeconomics I - Exam Reference

Contents

1 Notation and Definitions

Core Parameters and Variables

$$
\begin{aligned}
&q = \text{step size (quality jump, } q \geq \alpha^{-1}) \\
&k(s) = \text{quality rung in sector } s \\
&N = \text{number of products (fixed in QL, endogenous in Romer)} \\
&\tilde{X}_t(s) = q^{k(s)} X_t(s) = \text{quality-adjusted input} \\
&Q_t = \int_0^N q^{k(s)\frac{\alpha}{1-\alpha}} ds = \text{aggregate quality index} \\
&p = \text{innovation probability (Poisson arrival rate)} \\
&\xi = \text{R\&D cost parameter} \\
&\overline{\pi} = \frac{1-\alpha}{\alpha} \alpha^{\frac{2}{1-\alpha}} L = \text{baseline profits} \\
&\pi(k) = \overline{\pi} q^{\frac{\alpha}{1-\alpha}k} = \text{monopoly profits at rung } k
\end{aligned}
$$

Key Distinctions

Feature	**Quality Ladder**	**Expanding Varieties**
Innovation type	Vertical (quality improvement)	Horizontal (new products)
N	Fixed	Endogenous
Growth engine	Creative destruction	Love for variety
$\gamma^* \lesseqgtr \gamma$	Ambiguous	$\gamma^* > \gamma$ (too little)

2 Model Setup: Final Good Production

Final Good Production Function

$$Y_t = L^{1-\alpha} \int_0^N \tilde{X}_t^{\alpha}(s)\, ds \quad \text{where} \quad \tilde{X}_t(s) = q^{k(s)} X_t(s) \tag{Final Good}$$

Vertical vs Horizontal Differentiation

- **Vertical (Quality Ladder):** Same products improve in quality $q^{k(s)}$
- **Horizontal (Expanding Varieties):** New different products added to N

Key Assumption: $q \geq \alpha^{-1}$

This ensures monopolist prices above marginal cost, making innovation profitable.

Quality-Adjusted Input

$$\tilde{X}_t(s) = q^{k(s)} X_t(s)$$

Interpretation: A product at rung k is q^k times more productive than the baseline.

Quality Index

$$Q_t = \int_0^N q^{k(s)\frac{\alpha}{1-\alpha}} ds$$

Output with quality index:

$$Y_t = L^{1-\alpha} \overline{X}^{\alpha} Q_t = \alpha^{\frac{2\alpha}{1-\alpha}} L Q_t$$

3 Research Firms and Monopoly Profits

Monopoly Pricing

Firms set price equal to markup over marginal cost ($mc = 1$):

Monopoly Profits

Derivation of $\pi(k)$ Firm with quality k maximizes:

$$\pi = [P(s) - 1]X(s) \quad \text{s.t.} \quad X(s) = L\left(\frac{\alpha q^{\alpha k}}{P(s)}\right)^{\frac{1}{1-\alpha}}$$

Substituting $P = 1/\alpha$:

$$\pi(k) = \frac{1-\alpha}{\alpha}\alpha^{\frac{2}{1-\alpha}} L q^{\frac{\alpha}{1-\alpha}k} = \overline{\pi} q^{\frac{\alpha}{1-\alpha}k}$$

Poisson Innovation Process

Innovation arrives with probability $p\,dt$ in time dt. If innovation occurs, old firm gets zero profits thereafter.

Firm Value

Derivation of V With constant interest rate r and innovation probability p, the expected firm value satisfies:

$$rV = \pi(k) + p \cdot 0 - pV \quad \Rightarrow \quad V = \frac{\pi(k)}{r+p}$$

R&D Investment and Free Entry

Derivation of Innovation Probability R&D cost: $Z = \xi p q^{\frac{\alpha}{1-\alpha}(k+1)}$
Free entry condition (V = cost of innovation):

$$\frac{\overline{\pi} q^{\frac{\alpha}{1-\alpha}(k+1)}}{(r+p)\xi q^{\frac{\alpha}{1-\alpha}(k+1)}} = 1 \quad \Rightarrow \quad p = \frac{\overline{\pi}}{\xi} - r$$

Requires $\overline{\pi}/\xi > r$ for interior solution (positive innovation).

4 Growth Derivation

Quality Index Dynamics

Derivation of $\dot{Q}/Q$ In sector s, quality $q^{k(s)}$ stays same with prob $1-p$, jumps to $q^{k(s)+1}$ with prob p:

$$\mathbb{E}[\Delta Q] = \int_0^N p\left[q^{(k(s)+1)\frac{\alpha}{1-\alpha}} - q^{k(s)\frac{\alpha}{1-\alpha}}\right] ds = p(q^{\frac{\alpha}{1-\alpha}} - 1)Q$$

Balanced Growth Path Conditions

On BGP, all variables grow at same rate:

$$\gamma = \frac{\dot{C}}{C} = \frac{\dot{Q}}{Q} = \frac{\dot{Y}}{Y} = \frac{\dot{Z}}{Z} = \frac{\dot{X}}{X} \tag{BGP}$$

Market Growth Rate

Intuition: Growth comes from quality improvements (vertical innovation), not new varieties.

5 Market Equilibrium

Euler Equation

Households with CRRA utility (σ is IES):

$$\frac{\dot{c}_t}{c_t} = \frac{1}{\sigma}(r_t - \rho) \tag{EE}$$

Interest Rate

From free entry in R&D:

$$r = \frac{\overline{\pi}}{\xi} - p \tag{Interest Rate}$$

Solving for Equilibrium p and γ

Derivation From Euler: $\gamma = \frac{1}{\sigma}(\frac{\overline{\pi}}{\xi} - p - \rho)$
From quality growth: $\gamma = p(q^{\frac{\alpha}{1-\alpha}} - 1)$
Set equal and solve for p:

$$p\left[q^{\frac{\alpha}{1-\alpha}} - 1\right] = \frac{1}{\sigma}\left(\frac{\overline{\pi}}{\xi} - \rho\right) - \frac{p}{\sigma}$$

$$p\left[q^{\frac{\alpha}{1-\alpha}} - 1 + \frac{1}{\sigma}\right] = \frac{1}{\sigma}\left(\frac{\overline{\pi}}{\xi} - \rho\right)$$

For $\sigma = 1$ (log utility):

$$p = \left[\frac{\overline{\pi}}{\xi} - \rho\right] q^{-\frac{\alpha}{1-\alpha}}$$

6 Planner Solution

Planner vs Market Allocation

The planner sets price equal to marginal cost ($P^* = 1$), while market sets $P = 1/\alpha > 1$.

Derivation of X^* Market FOC: $1/\alpha = \alpha L^{1-\alpha} q^{\alpha k} X^{\alpha-1} \Rightarrow X = L\alpha^{\frac{2}{1-\alpha}} q^{\frac{\alpha k}{1-\alpha}}$
Planner FOC: $1 = \alpha L^{1-\alpha} q^{\alpha k} X^{*\alpha-1} \Rightarrow X^* = L\alpha^{\frac{1}{1-\alpha}} q^{\frac{\alpha k}{1-\alpha}}$

$$X^* = L\alpha^{\frac{1}{1-\alpha}} q^{\frac{\alpha k}{1-\alpha}} \quad > \quad X = L\alpha^{\frac{2}{1-\alpha}} q^{\frac{\alpha k}{1-\alpha}} \qquad \text{(Input Comparison)}$$

Since $\alpha^{\frac{1}{1-\alpha}} > \alpha^{\frac{2}{1-\alpha}}$, we have $X^* > X$.

Output Comparison

$$Y = L\alpha^{\frac{2\alpha}{1-\alpha}} Q \quad < \quad Y^* = L\alpha^{\frac{\alpha}{1-\alpha}} Q \qquad \text{(Output Comparison)}$$

The planner achieves higher output for given Q because of lower markup (no monopoly distortion).

Planner Growth Rate

Derivation of γ^* Planner interest rate (implicit):

$$r^* = \frac{\pi^*}{\xi}\left[1 - q^{-\frac{\alpha}{1-\alpha}}\right]$$

where $\pi^* = L\frac{1-\alpha}{\alpha}\alpha^{\frac{1}{1-\alpha}} > \overline{\pi}$
Planner growth:

$$\gamma^* = \frac{1}{\sigma}\left[r^* - \rho\right]$$

Unlike Expanding Varieties where market always underinvests, QL has ambiguous welfare result due to competing externalities.

7 Externalities and Inefficiency

Two Conflicting Externalities

Business Stealing Effect	**Temporary Returns Effect**
Entrants don't internalize that they destroy incumbent profits	Present innovators ignore that future innovators benefit from their innovation
Leads to **too much innovation**	Leads to **too little innovation**
Negative externality	Positive externality

Mathematical Representation

- Business stealing: $\rho q^{-\frac{\alpha}{1-\alpha}} > 0$ added to interest rate
- This term represents extraction of monopoly profit from predecessor

Planner vs Market: $\gamma^* \lesseqgtr \gamma$

$$\pi^* = L\frac{1-\alpha}{\alpha}\alpha^{\frac{1}{1-\alpha}} > \overline{\pi} = L\frac{1-\alpha}{\alpha}\alpha^{\frac{2}{1-\alpha}}$$

Planner values innovation more than individual inventors.

Comparison with Other Models

Model	**Externality**	**Welfare**
Lab-Equipment	Positive (knowledge spillover)	$\gamma^* > \gamma$ (too little)
Expanding Varieties	Positive (love for variety)	$\gamma^* > \gamma$ (too little)
Quality Ladder	Both positive and negative	$\gamma^* \lesseqgtr \gamma$ (ambiguous)

8 Competition and Innovation (Aghion et al., 2005)

Neck-and-Neck vs Unlevel Sectors

Neck-and-Neck	**Unlevel**
Both firms at same quality rung	One firm ahead (leader)
Competition reduces profits ($\pi_0 < \pi_1$)	Leader earns π_1, follower earns 0
Escape competition effect dominates	Schumpeterian effect dominates

Innovation Incentives

- **Unlevel sector:** Laggard innovates to catch up; leader has no incentive
- **Neck-and-neck sector:** Both innovate to escape competition and get monopoly profits

U-Shaped Relationship

Derivation of Innovation-Competition Relation Let $\Delta \in [0, 1]$ parameterize competition ($\Delta \to 1$: competition, $\Delta \to 0$: collusion).
Innovation in unlevel: $n_{-1} = (1 - \Delta)\frac{\pi_1}{\psi}$
Innovation in neck-and-neck: $n_0 = \frac{\Delta \pi_1}{\psi}$
Steady-state innovation rate:

$$I = \frac{2\pi_1}{\psi(1 + \psi h/\pi_1)} \Delta \left(1 - \Delta + \frac{\psi h}{\pi_1}\right)$$

Interpretation

- **Low competition ($\Delta < 1/2$):** Escape competition effect dominates $\Rightarrow$ more competition $\Rightarrow$ more innovation
- **High competition ($\Delta > 1/2$):** Schumpeterian effect dominates $\Rightarrow$ more competition $\Rightarrow$ less innovation

Empirical Prediction

U-shaped relationship between competition and innovation (Aghion et al., 2005).

9 Policy Analysis

R&D Tax Effects

Consider a tax τ on R&D investment. The free entry condition becomes:

$$\frac{V(k+1)}{\xi q^{\frac{\alpha}{1-\alpha}(k+1)}} = 1 + \tau$$

Derivation of $p(\tau)$ With tax, R&D cost increases. For free entry to hold, expected benefit must rise:

$$r(\tau) = \frac{\pi q^{-\frac{\alpha}{1-\alpha}(k+1)}}{\xi(1+\tau)} - p(\tau)$$

Since $p(\tau)$ decreases with τ (higher cost lowers innovation), and r falls:

$$\gamma(\tau) = \frac{1}{\sigma}[r(\tau) - \rho] \quad \Rightarrow \quad \frac{\partial \gamma}{\partial \tau} < 0$$

Incumbent Lobbying

Current monopolists benefit from taxing R&D because it reduces future competition. They may lobby for such policies.

If R&D is banned ($\tau \to \infty$):

- Current monopolists' value increases significantly
- No threat of creative destruction
- Innovators would innovate then lobby against further innovation

10 Problem-Solving Templates

(a) Derive Growth Rate in Quality Ladder Model

1. Start from quality index: $Q = \int_0^N q^{k(s)\frac{\alpha}{1-\alpha}} ds$
2. Compute expected change: $\mathbb{E}[\Delta Q] = p(q^{\frac{\alpha}{1-\alpha}} - 1)Q$
3. Take limit: $\dot{Q}/Q = p(q^{\frac{\alpha}{1-\alpha}} - 1)$
4. Use Euler equation: $\gamma = \frac{1}{\sigma}(\frac{\overline{\pi}}{\xi} - p - \rho)$
5. Solve two equations for p and γ

(b) Compare Market vs Planner

1. Market: $P = 1/\alpha$, Planner: $P^* = 1$
2. Derive X vs X^*: $X^* > X$ since $\alpha^{\frac{1}{1-\alpha}} > \alpha^{\frac{2}{1-\alpha}}$
3. Compare outputs: $Y^* > Y$
4. Compare growth: analyze externalities (business stealing vs temporary returns)
5. Conclude: $\gamma^* \lesseqgtr \gamma$ (ambiguous)

(c) Analyze Competition-Innovation U-Shape

1. Distinguish neck-and-neck vs unlevel sectors
2. In neck-and-neck: escape competition effect $\Rightarrow$ more competition $\Rightarrow$ more innovation
3. In unlevel: Schumpeterian effect $\Rightarrow$ more competition $\Rightarrow$ less innovation
4. Aggregate: $\partial I/\partial \Delta = 1 - 2\Delta + \psi h/\pi_1$
5. Result: U-shaped relationship

(d) Solve for Equilibrium Innovation Probability

1. Write firm value: $V = \pi(k)/(r + p)$
2. Write R&D cost: $Z = \xi p q^{\frac{\alpha}{1-\alpha}(k+1)}$
3. Free entry: $V = Z$
4. Solve: $p = \overline{\pi}/\xi - r$
5. Combine with Euler: solve system for p and γ

11 Common Pitfalls

1. **Confusing QL with Romer (Expanding Varieties)**
   - QL: Vertical innovation, fixed N, creative destruction
   - Romer: Horizontal innovation, endogenous N, no destruction
2. **Forgetting Creative Destruction**
   - In QL, innovation destroys incumbent profits
   - This is the key mechanism for Schumpeterian growth
3. **Misinterpreting Business Stealing**
   - Business stealing = entrants don't internalize destroying incumbents
   - Leads to too much innovation (not too little!)
4. **Confusing π^* vs $\overline{\pi}$**
   - π^*: planner's implicit profits $= L\frac{1-\alpha}{\alpha}\alpha^{\frac{1}{1-\alpha}}$
   - $\overline{\pi}$: market baseline profits $= L\frac{1-\alpha}{\alpha}\alpha^{\frac{2}{1-\alpha}}$
   - $\pi^* > \overline{\pi}$: planner values innovation more
5. **Misunderstanding Scale Effects**
   - QL: Growth depends on $\overline{\pi}/\xi - \rho$, which depends on L
   - Scale effects present (unlike semi-endogenous models)
6. **Assuming $\gamma^* > \gamma$ always**
   - Only true for Expanding Varieties, not for QL
   - QL has ambiguous welfare result

12 Quick Reference Formulas

Core Equations

$$\tilde{X} = q^k X \qquad \text{(Quality-Adjusted Input)}$$

$$Q = \int_0^N q^{k(s)\frac{\alpha}{1-\alpha}} ds \qquad \text{(Quality Index)}$$

$$\pi(k) = \overline{\pi}\, q^{\frac{\alpha}{1-\alpha}k} \qquad \text{(Monopoly Profits)}$$

$$V = \frac{\pi(k)}{r+p} \qquad \text{(Firm Value)}$$

$$p = \frac{\overline{\pi}}{\xi} - r \qquad \text{(Innovation Probability)}$$

Growth Equations

$$\frac{\dot{Q}}{Q} = p\left[q^{\frac{\alpha}{1-\alpha}} - 1\right] \qquad \text{(Quality Growth)}$$

$$\gamma = \left[\frac{\overline{\pi}}{\xi} - \rho\right]\left[1 - q^{-\frac{\alpha}{1-\alpha}}\right] \qquad \text{(Market Growth)}$$

$$r = \frac{\overline{\pi}}{\xi}\left[1 - q^{-\frac{\alpha}{1-\alpha}}\right] + \rho q^{-\frac{\alpha}{1-\alpha}} \qquad \text{(Interest Rate)}$$

Planner vs Market

$$X^* > X \qquad \text{(Input)}$$

$$Y^* > Y \qquad \text{(Output)}$$

$$\gamma^* \lesseqgtr \gamma \qquad \text{(Growth (ambiguous))}$$

Competition-Innovation

$$\frac{\partial I}{\partial \Delta} \propto 1 - 2\Delta + \frac{\psi h}{\pi_1} \qquad \text{(U-Shape)}$$

Key Parameters

$$\overline{\pi} = \frac{1-\alpha}{\alpha}\alpha^{\frac{2}{1-\alpha}} L$$

$$\pi^* = \frac{1-\alpha}{\alpha}\alpha^{\frac{1}{1-\alpha}} L$$

13 Implementation Notes

Key Differences from Expanding Varieties

1. Growth engine: Quality improvements (vertical) vs new varieties (horizontal) N fixed vs endogenous Creative destruction vs no destruction $\gamma^* \lesseqgtr \gamma$ vs $\gamma^* > \gamma$

Exam Tips

- Always distinguish vertical from horizontal differentiation
- Remember the two externalities in QL: business stealing (too much) and temporary returns (too little)
- Know the U-shaped competition-innovation relationship
- Don't assume market underinvests in R&D (unlike Romer!)

Compilation

Compile with: `pdflatex` or `lualatex`